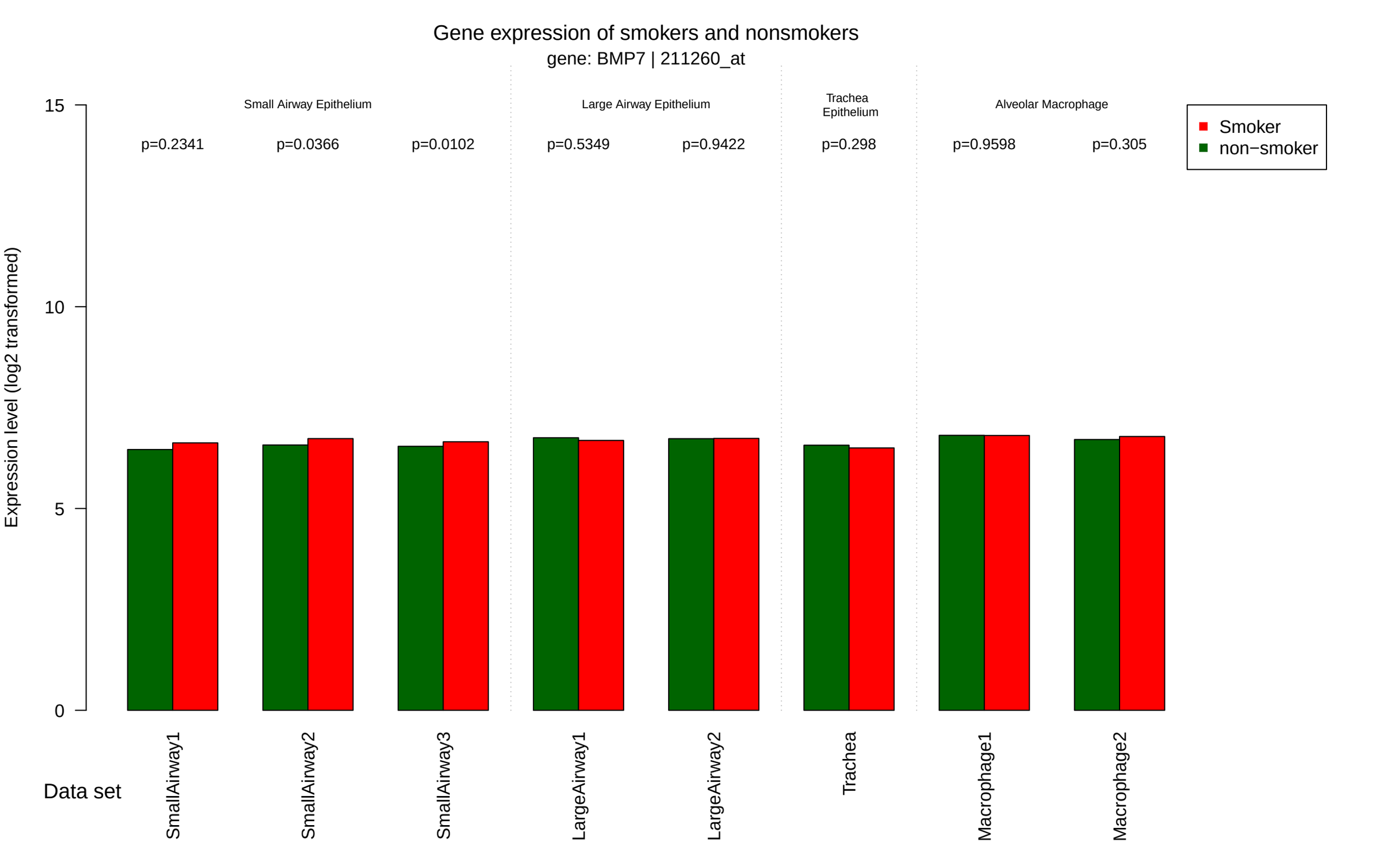
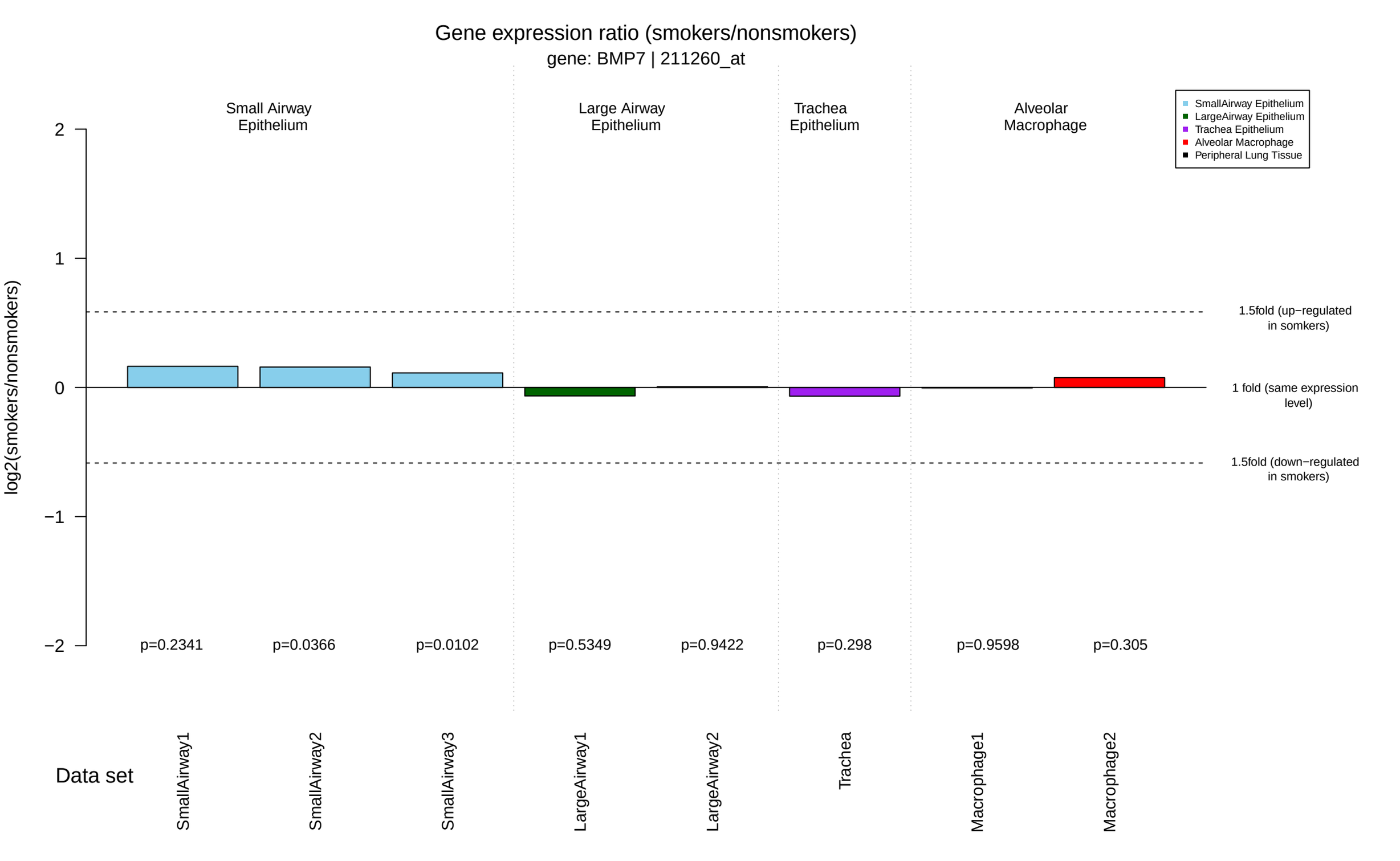
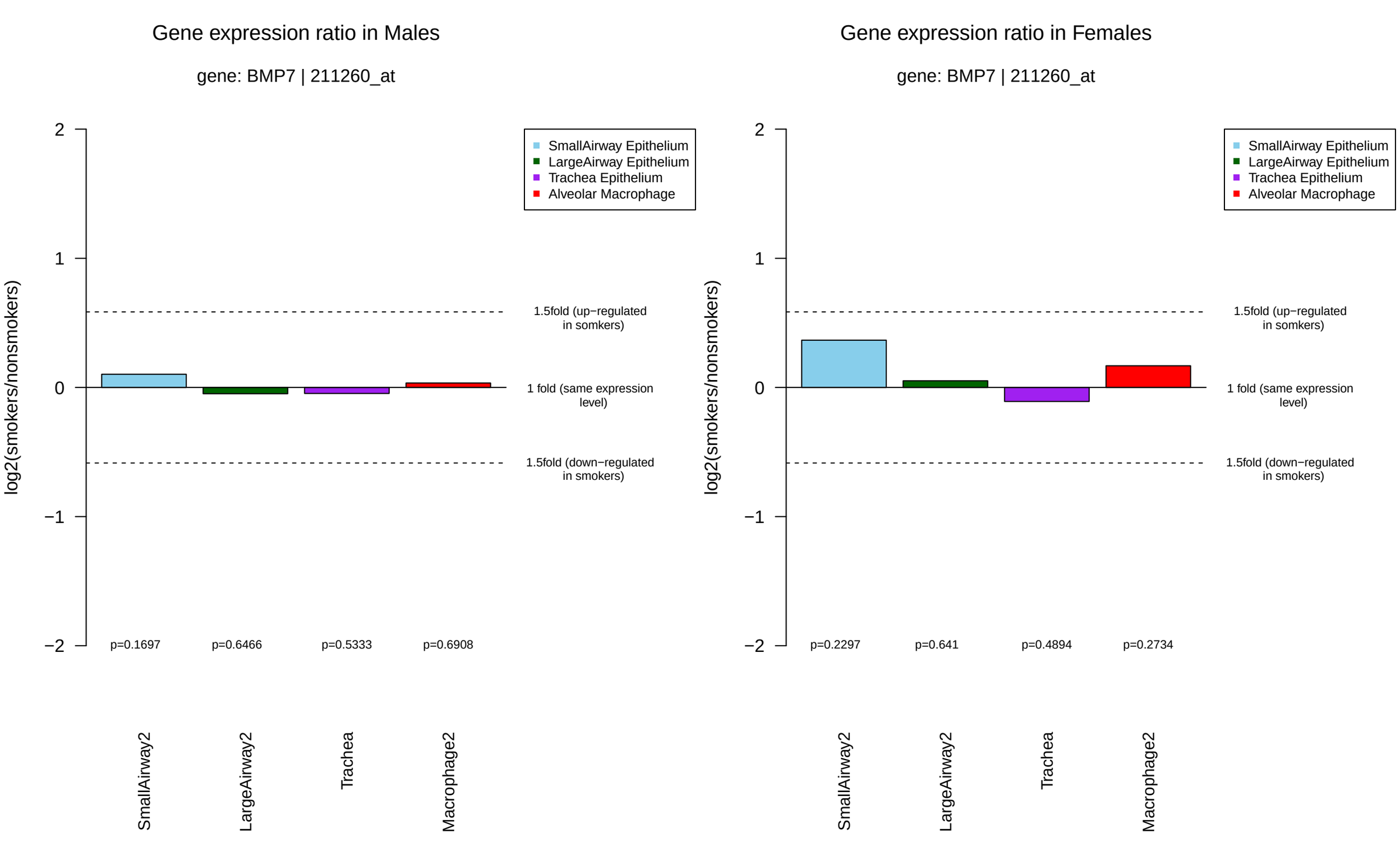
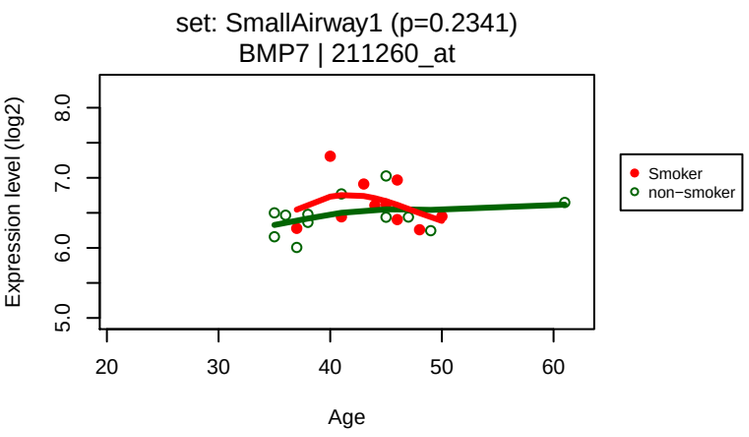
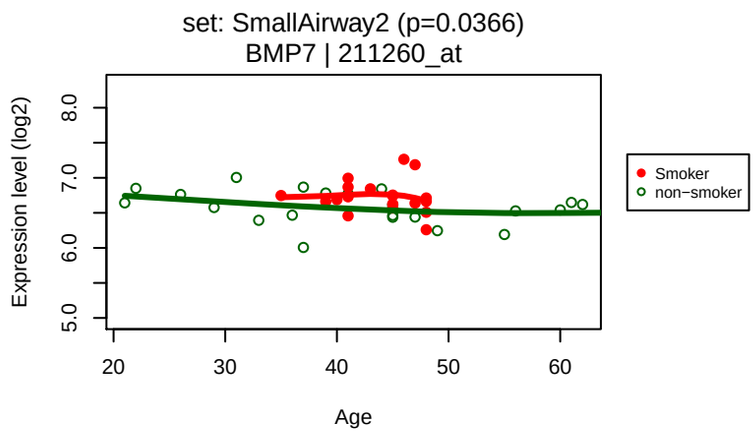
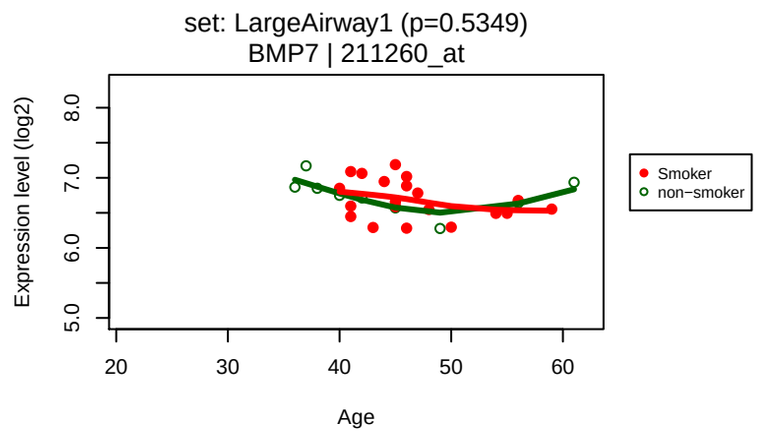

Gene expression of smokers and nonsmokers
gene: BMP7 | 211260_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2341
p=0.0366
p=0.0102
p=0.5349
p=0.9422
p=0.298
p=0.9598
p=0.305
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

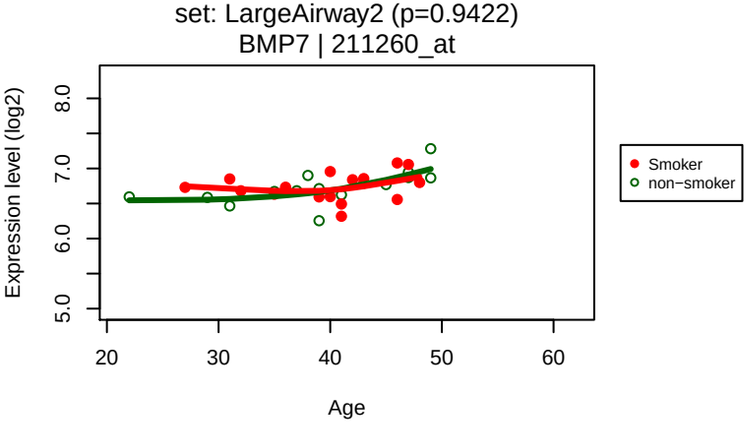

Gene expression ratio (smokers/nonsmokers)
gene: BMP7 | 211260_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2341
p=0.0366
p=0.0102
p=0.5349
p=0.9422
p=0.298
p=0.9598
p=0.305
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

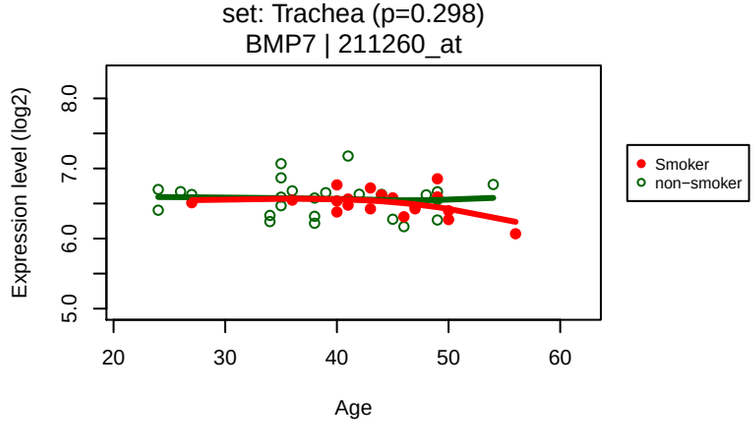

Gene expression ratio in Males
gene: BMP7 | 211260_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1697
p=0.6466
p=0.5333
p=0.6908
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP7 | 211260_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2297
p=0.641
p=0.4894
p=0.2734
SmallAirway2
LargeAirway2
Trachea
Macrophage2

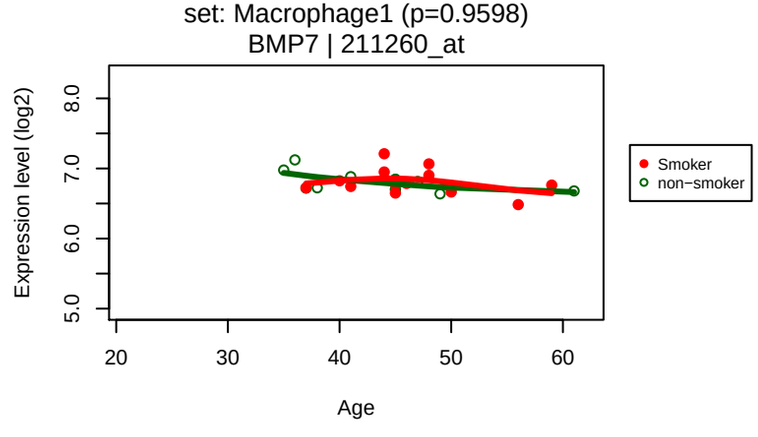

set: SmallAirway1 (p=0.2341)
BMP7 | 211260_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker

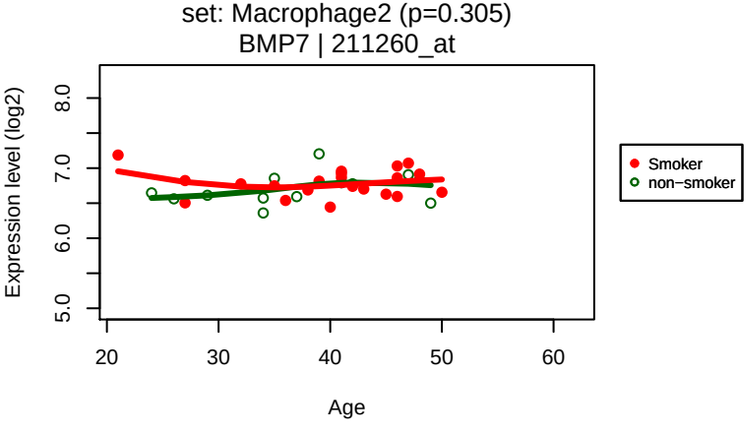

set: SmallAirway2 (p=0.0366)
BMP7 | 211260_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker

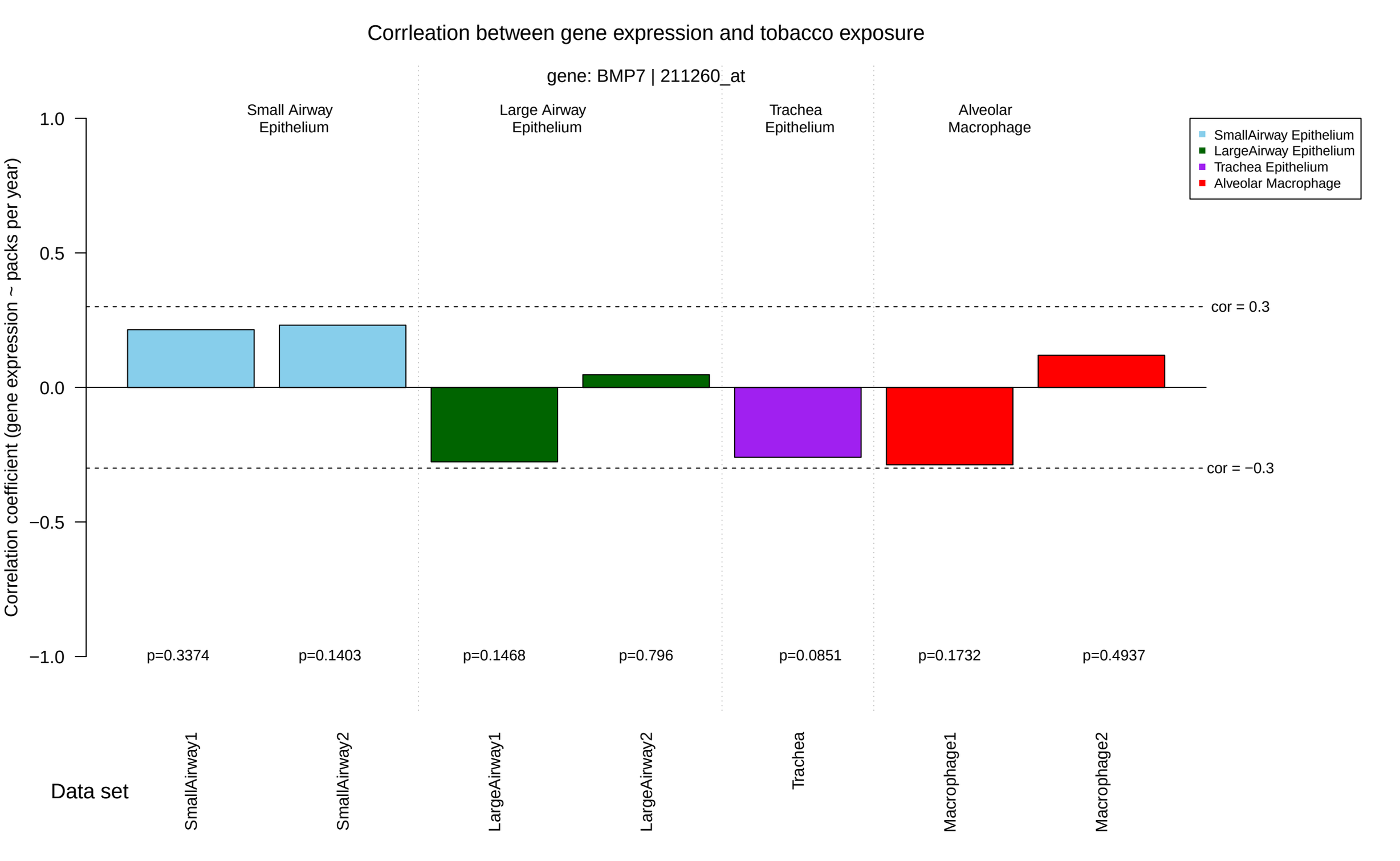

set: LargeAirway1 (p=0.5349)
BMP7 | 211260_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9422)
BMP7 | 211260_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.298)
BMP7 | 211260_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9598)
BMP7 | 211260_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.305)
BMP7 | 211260_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP7 | 211260_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3374
p=0.1403
p=0.1468
p=0.796
p=0.0851
p=0.1732
p=0.4937
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2